Differentiation: Connected Rates of Change

Questions 1–16

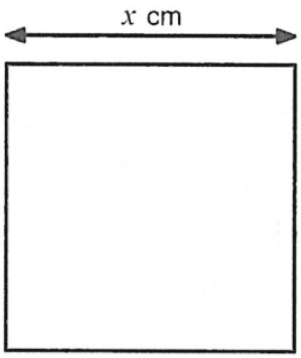
Question 1. State each of the following:

a) The area of a circle in terms of its radius r
b) The area of an equilateral triangle in terms of its side length x
c) The volume of a cylinder V in terms of its radius r and its height h
d) The volume of a sphere in terms of its radius r
e) The surface area of a sphere in terms of its radius r
f) The volume of a cone with base of radius r and height h in terms of r and h

Question 2. Given

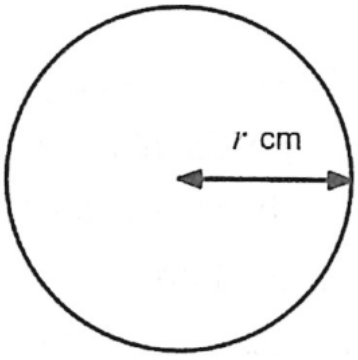
a) $A = \frac{1}{2}\pi r^2$ and $\frac{dr}{dt} = 2$, find $\frac{dA}{dt}$ in terms of r
b) $y = 10x^2$ and $\frac{dx}{dt} = -\frac{1}{2}$, find $\frac{dy}{dt}$ in terms of x
c) $P = 2x^3 - x$ and $\frac{dx}{dt} = 3$, find $\frac{dP}{dt}$ in terms of x
d) $M = -2x^5 + x^2$ and $\frac{dx}{dt} = \frac{1}{2}$, find $\frac{dM}{dt}$ in terms of x

Question 3. The side of a square x is increasing at a rate of 10 $\mathrm{cm\,s^{-1}}$.



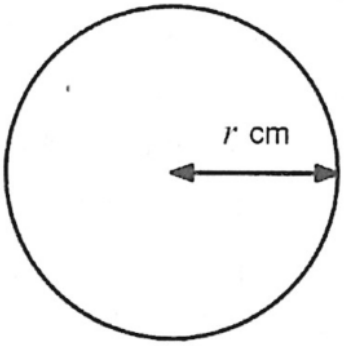
Find the rate at which the area A of the square is increasing when the side of the square is exactly 4 cm.

Question 4. The radius of a circle r is increasing at a rate of 3 $\mathrm{cm\,s^{-1}}$.



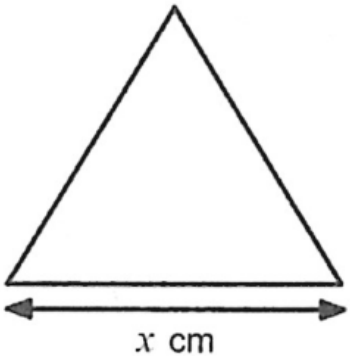
Find the rate at which the area A of the circle is increasing when the radius of the circle is 5 cm giving your answer in terms of π

Question 5. The radius of a circle r is decreasing at a constant rate of $0.1\ \text{cm}\,\text{s}^{-1}$.



Find the rate at which the perimeter of the circle is decreasing, giving your answer in terms of π

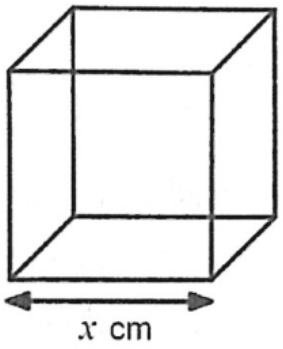
Question 6. The side of an equilateral triangle x is increasing at a rate of $12\ \text{cm}\,\text{s}^{-1}$.



Find the rate at which the area of the triangle is increasing when the side of the triangle is $2\sqrt{3}$ cm.

Question 7. The surface area of a sphere is increasing at a rate of 12π $\text{cm}^2\,\text{s}^{-1}$. Find the rate of change of the radius of the sphere exactly when its diameter is 6 cm.

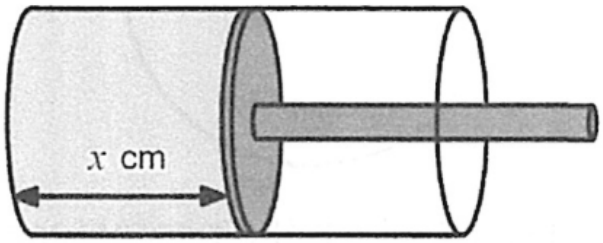
Question 8. A cube is growing so that its volume increases at a rate of 5 $\text{cm}^3\,\text{s}^{-1}$.



The length of each of its sides is x cm.

a) State the volume of the cube V and the surface area of the cube A in terms of x
b) Show that the rate of increase of the surface area of the cube may be given by $\frac{dA}{dt} = \frac{20}{x}$
c) Hence, find the rate of increase of the cube's surface area when x has a length of 4 cm.

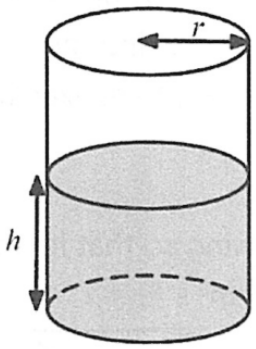
Question 9. A piston can slide inside a combustion cylinder which is closed at one end. The cylinder is filled with gas whose pressure p in atmospheric units is inversely proportional to the distance of the piston from the closed end, x cm.



It is known that the atmospheric pressure is 1.2 when x is 50 cm. At a different point in time, the distance from the closed end is 25 cm and the piston is moving away from the closed end at 5 cm per minute.
Find the rate at which the pressure is changing at that point in time.

Question 10. A sphere is growing so that its volume increases at a rate of 4 $\text{cm}^3\,\text{s}^{-1}$. Find the rate of increase of the sphere's surface area when its diameter is 50 cm.

Question 11. A cylindrical tank with radius r cm has water being poured into it at a rate of 10 $\text{cm}^3\,\text{s}^{-1}$.



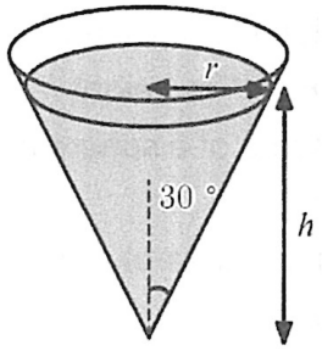
The height of water after t seconds is given by h cm.

a) Find an expression for the rate of increase of the water level in the tank in terms of r

b) Given the tank has diameter 90 cm, find the rate of change of the water level giving your answer to 3 significant figures.

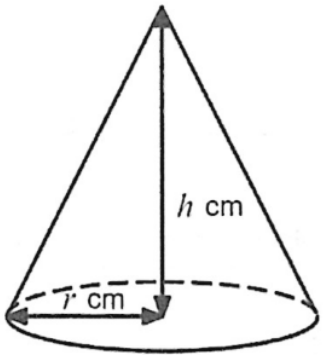
Question 12. A cylindrical tank with radius 10 cm which initially starts full, has water dripping out of a hole in the bottom at a rate of 5 cm^3 per minute. The height of water after t minutes is given by h cm.
Find the rate of change of the water level giving your answer to 3 significant figures.

Question 13. Water is being poured into an empty cylindrical cone of vertical height 10 cm at a constant rate of 6 $\text{cm}^3\,\text{s}^{-1}$. The cylindrical cone has an angle of 30° to the vertical.



a) Show that the volume of the water V when it is at height h cm is given by $V = \frac{1}{9}\pi h^3$
b) Find $\frac{dV}{dh}$
c) Find $\frac{dh}{dt}$ when the cone is full.

Question 14. Sand is being poured onto a table and forms a pile in the shape of a cone. The height of the cone is equal to the diameter of the base.



The sand is poured at 3 cm^3 per second. Find the rate at which the height of the pile is increasing when the base has a diameter of 4 cm.

Question 15. The volume of a cube with sides of length x metres is increasing at a constant rate.
Show that the rate of increase of the sides, is inversely proportional to the area of one side of the cube.

Question 16. The radius of a circle r changes at a rate which is inversely proportional to the square of r

a) Given that the rate is 0.5 $\text{cm}\,\text{s}^{-1}$ when the radius is 2 cm, find $\frac{dr}{dt}$

b) Show that the rate at which the area of the circle, A, changes satisfies $\frac{dA}{dt} = \sqrt{\frac{16\pi^3}{A}}$